

1-0893 Pressure Effects on the Magnetic Transitions in the Heavy-fermion Compounds YbTrGe (*Tr* = Ni, Pt, Rh, and Ir)



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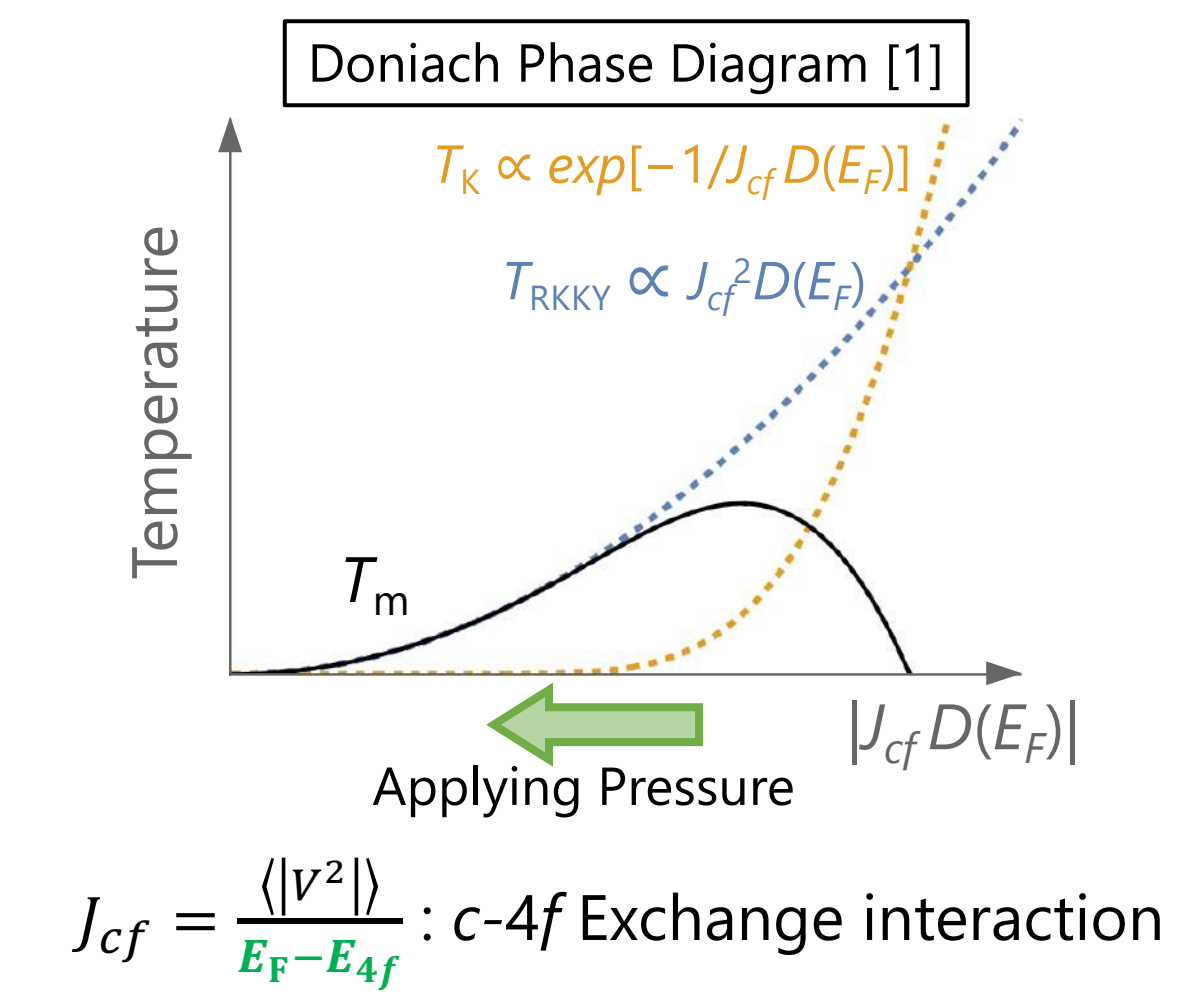
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Introduction

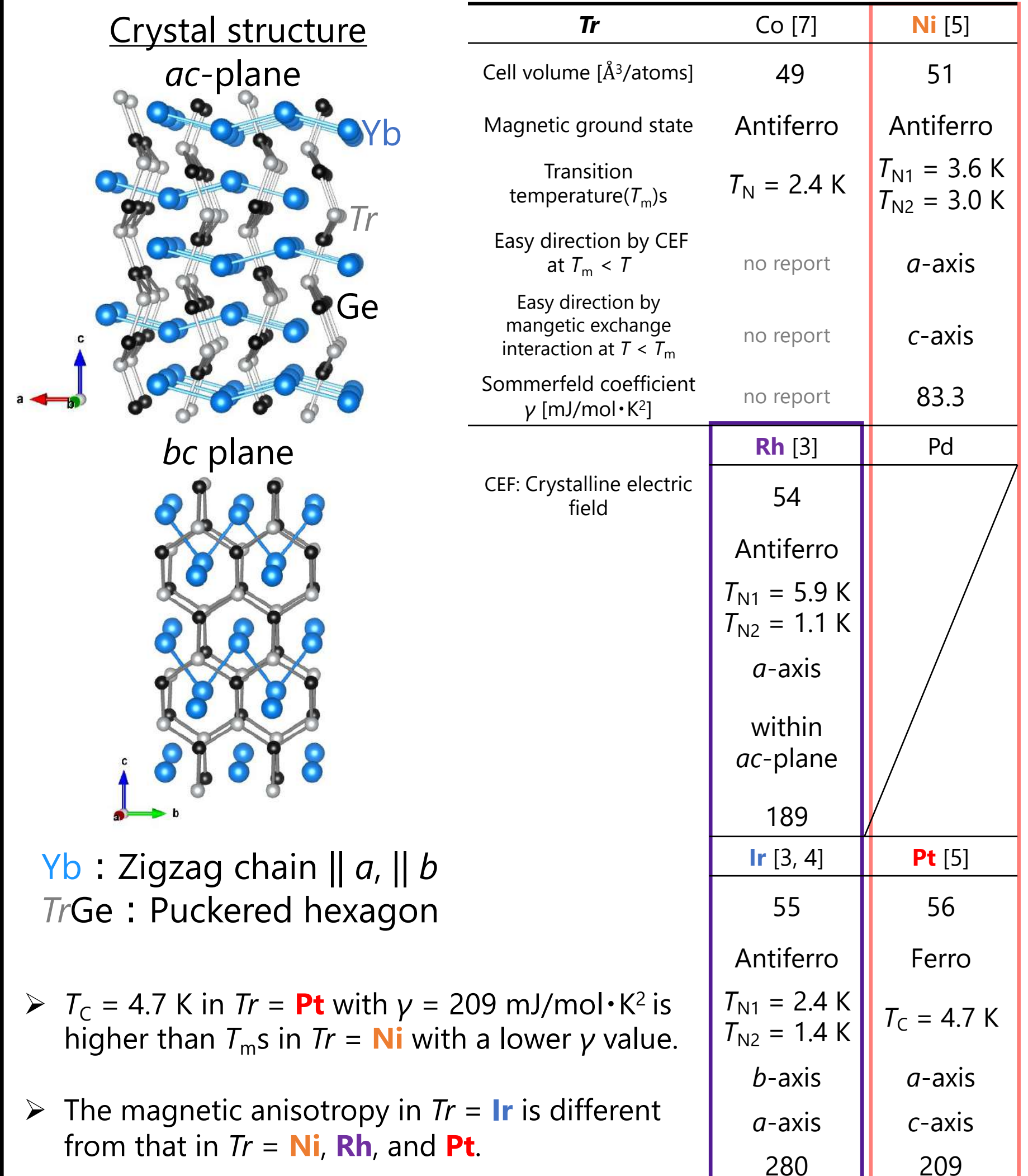
Pressure Effect on the magnetism of Yb compounds



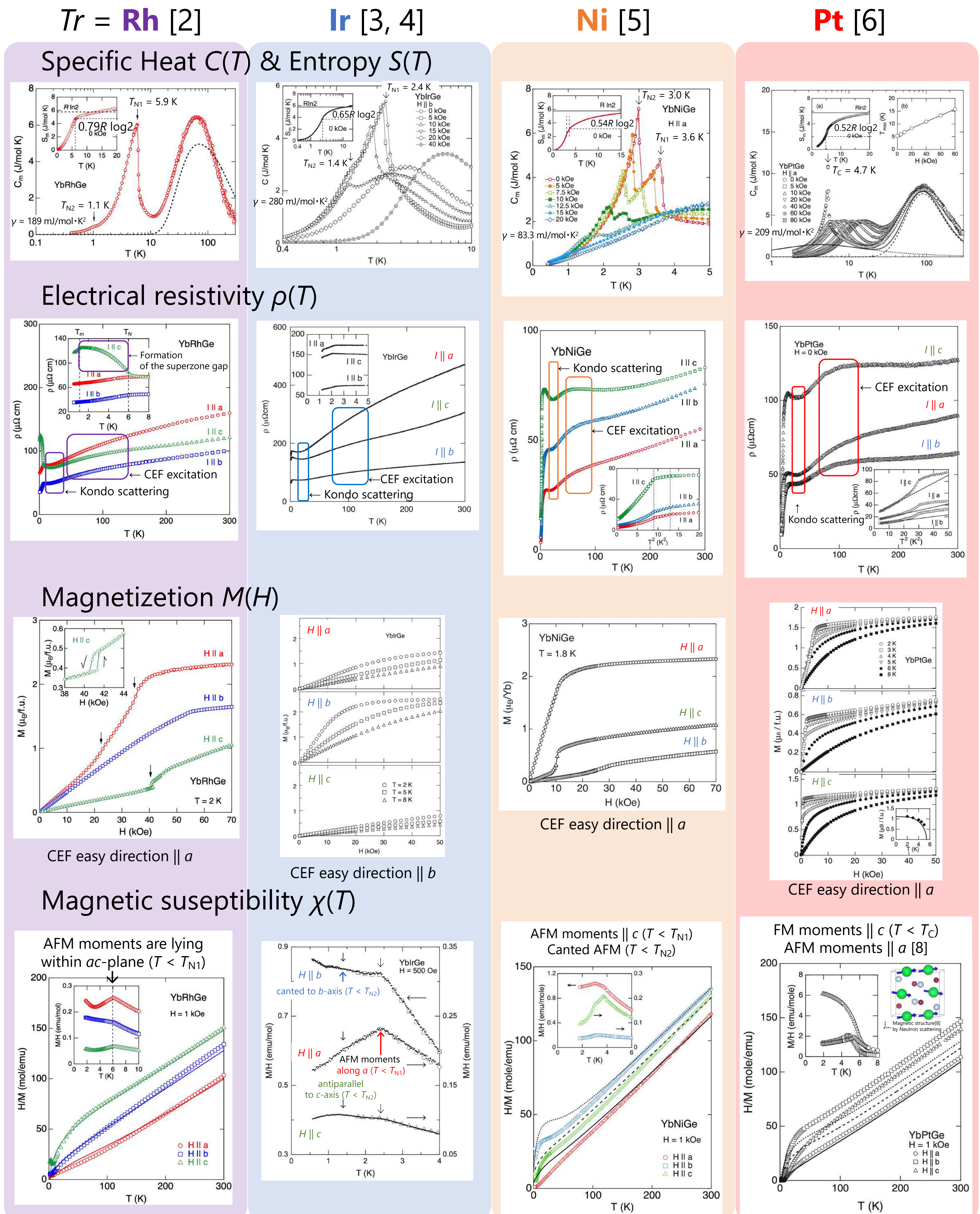
Pressure(*P*) effect on *Jcf*

- c-f* hole hybridization; $\langle |V^2| \rangle$: ↑ with *P*
- $E_F - E_{4f}$: ↑ with *P*
- $\Delta \langle |V^2| \rangle < \Delta |E_F - E_{4f}|$ with *P* because $\Delta \langle |V(\text{Ce})^2| \rangle > \Delta \langle |V(\text{Yb})^2| \rangle$ with *P*.
→ J_{cf} : ↓

YbTrGe with the orthorhombic ε-TiNiSi-type structure



Physical properties of YbTrGe at ambient pressure



Question

How is the magnetism of the isostructural series of YbTrGe (*Tr* = **Ni**, **Pt**, **Rh**, and **Ir**) explained by the Doniach phase diagram?

→ We have studied pressure dependence of *T*_ms by the measurement of electrical resistivity $\rho(T)$ at various constant pressures up to 2.5 GPa.

Experimental Methods

Samples

Single crystals of YbTrGe (*Tr* = **Ni**, **Pt**, **Rh**, and **Ir**) were grown by the Bridgmann method.

Electrical resistivity measurement

- AC-four terminal method
- Applying electrical current along the *c*-axis for *Tr* = **Ni**, **Pt**, and **Rh** and along the *b*-axis for *Tr* = **Ir**.

Pressurization

- Pressure cell: Piston cylinder cell (up to *P* = 2.5 GPa)
- Pressure transmitting medium: Daphne oil 7474

Evaluation of pressure at low temperature

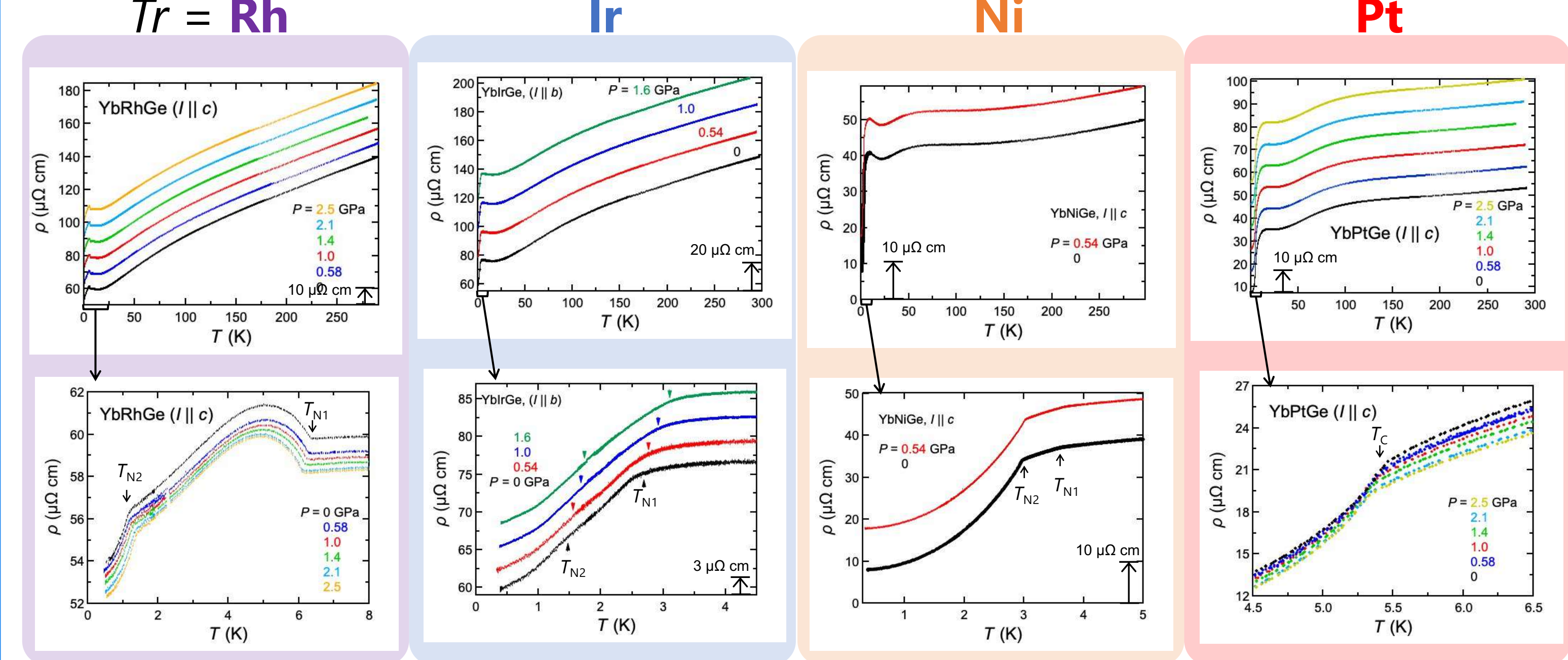
Variation of superconducting transition temperature *T*_C(*P*) of Sn
 $\Delta T_C(P) = 3.732 - 4.95 \times 10^{-5}P + 3.9 \times 10^{-10}P^2$ [unit of *P*: atm] [9]

Refrigerator

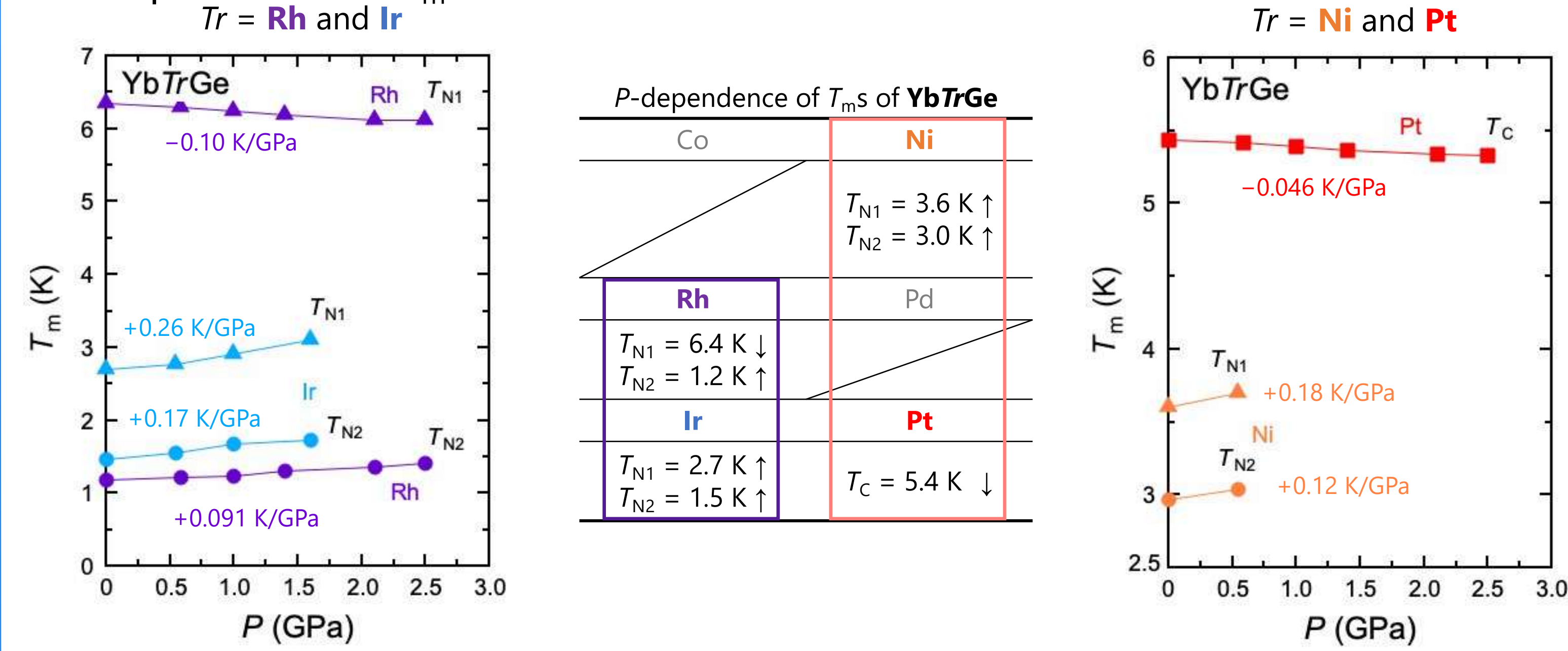
³He refrigerator, Heliox (0.3 < *T* < 300 K)

Results

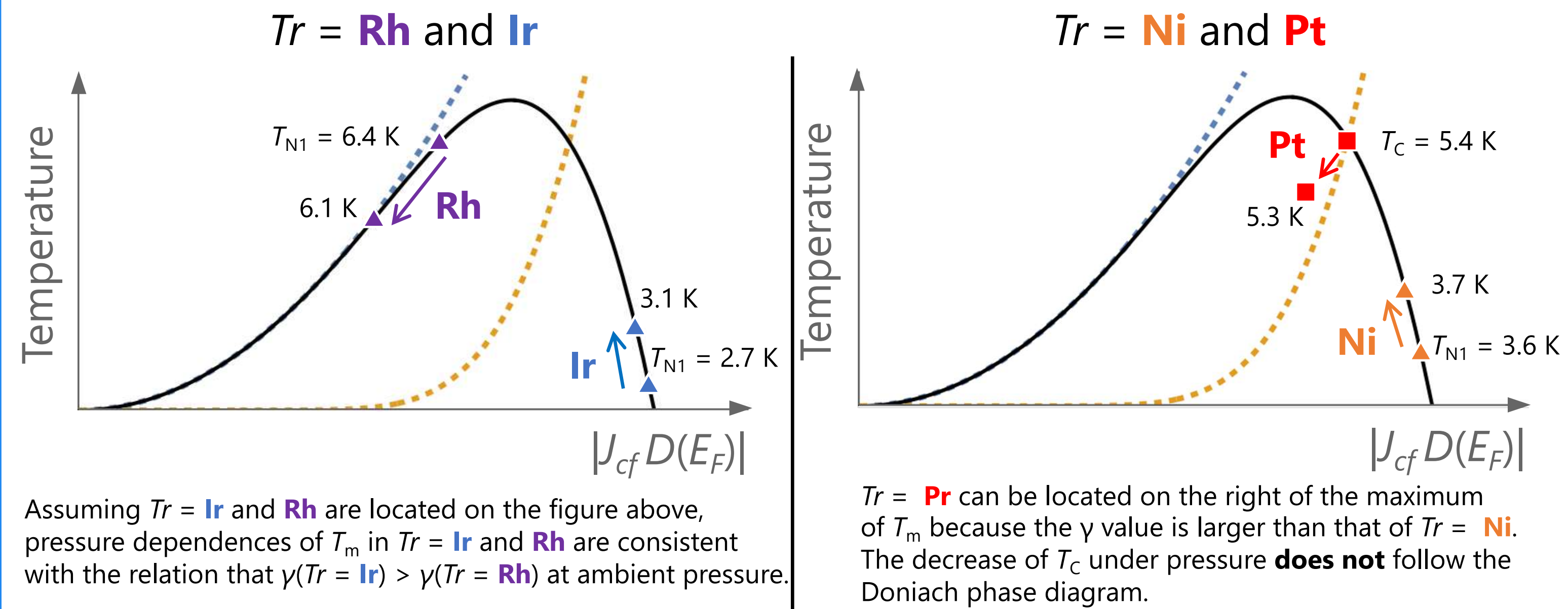
Resistivity $\rho(T)$ under pressure *P* up to 2.5 GPa



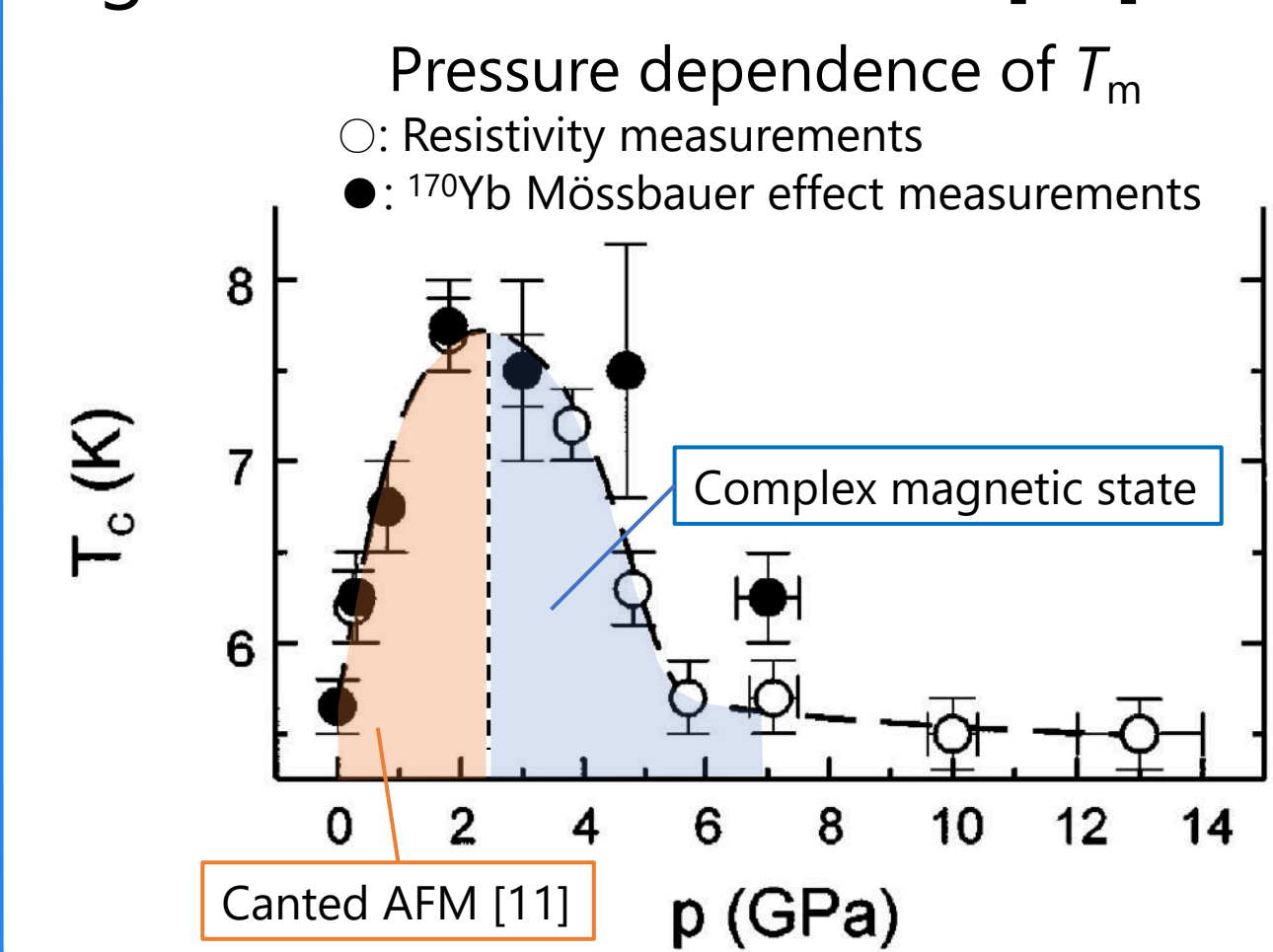
P dependence of *T*_ms in YbTrGe



Discussion of *T*_m(*P*) using the Doniach phase diagram

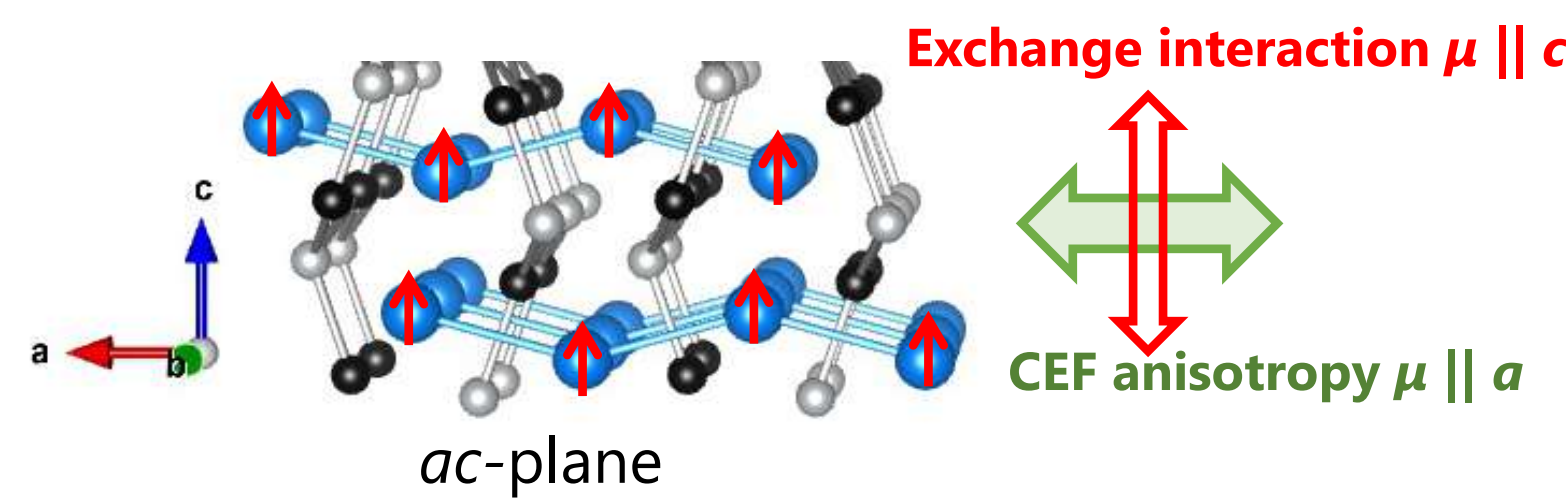


Pressure Effect on the magnetic ground state in YbNiSn [10]



The CEF effect and the exchange interaction may compete in *Tr* = **Pt**.

On cooling to *T*_C, the magnetic anisotropy || *c*-axis **overcomes** the CEF magnetic anisotropy || *a*-axis.



- Applying pressure enhances the CEF effect on a single Yb³⁺ site.

The increased the magnetic anisotropy along the *a*-axis disrupts the ferromagnetic order along the *c*-axis. → **Decreases** *T*_C

Summary

- We have studied pressure(*P*) dependences of *T*_ms in YbTrGe (*Tr* = **Ni**, **Pt**, **Rh**, and **Ir**) under pressures up to 2.5 GPa by resistivity measurements.
- P* dependences of *T*_ms in YbTrGe (*Tr* = **Ni**, **Rh**, and **Ir**) are explained by the Doniach phase diagram.
- The unexpected suppression of *T*_C in Yb**Pt**Ge is explained by a competition between the CEF anisotropy(|| *a*) and the intersite exchange interaction anisotropy(|| *c*).

References

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